

$$1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots < 2^n < 3^n < \dots < n^n$$

Asymptotic Notations

O big-oh upper bound

Ω big-omega Lower bound

$\rightarrow \Theta$ theta Average bound

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Big-oh

The function $f(n) = O(g(n))$ iff $\exists$ +ve constants c and n_0

such that $f(n) \leq c * g(n) \forall n \geq n_0$

eg: $f(n) = 2n + 3$

$$2n + 3 \leq \underbrace{10}_{c} \underbrace{n}_{g(n)} \quad n \geq 1$$

$\uparrow$ $\uparrow$ $\uparrow$
 $f(n)$ c $g(n)$

$$\therefore f(n) = O(n)$$

Properties of Asymptotic Notations

General Properties

- if $f(n)$ is $O(g(n))$ then $a \cdot f(n)$ is $O(g(n))$

eg: $f(n) = 2n^2 + 5$ is $O(n^2)$

then $7 \cdot f(n) = 7(2n^2 + 5)$

$= 14n^2 + 35$ is $O(n^2)$

Properties of Asymptotic Notations

General Properties

- if $f(n)$ is $\Omega(g(n))$ then $a \cdot f(n)$ is $\Omega(g(n))$

eg: $f(n) = 2n^2 + 5$ is $O(n^2)$

$$\begin{aligned} \text{then } 7 \cdot f(n) &= 7(2n^2 + 5) \\ &= 14n^2 + 35 \text{ is } O(n^2) \end{aligned}$$

Properties of Asymptotic Notations

Reflexive

if $f(n)$ is given then $f(n) = O(f(n))$

$$\text{eg: } f(n) = n^2 \quad O(\underset{\uparrow}{n^2})$$

Properties of Asymptotic Notations

Transitive

if $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$

then $f(n) = O(h(n))$

eg: $f(n) = n$ $g(n) = n^2$ $h(n) = n^3$

n is $O(n^2)$ and n^2 is $O(n^3)$

then n is $O(n^3)$

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General Properties

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if $f(n)$ is given then $f(n)$ is $O(f(n))$

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n is $O(n^2)$ and n^2 is $O(n^3)$

then n is $O(n^3)$

Properties of Asymptotic Notations

Symmetric

if $f(n)$ is $\Theta(g(n))$ then $g(n)$ is $\Theta(f(n))$

eg: $f(n) = n^2$ $g(n) = n^2$

$$f(n) = \Theta(n^2)$$

$$g(n) = \Theta(n^2)$$

Properties of Asymptotic Notations

Transpose Symmetric

if $f(n) = O(g(n))$ then $g(n)$ is $\Omega(f(n))$

eg: $f(n) = n$ $g(n) = n^2$

then n is $O(n^2)$ and
 n^2 is $\Omega(n)$